Ages of the Duluth Gabbro Complex and of the Endion Sill, Duluth, Minnesota[1]

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INTRODUCTION

The Duluth gabbro complex is a large sill-like, composite intrusive consisting of mafic igneous rocks. It crops out within the city of Duluth, Minnesota, and extends for about 150 miles in a northeasterly direction west of Lake Superior. The Duluth gabbro was intruded along the base of the North Shore volcanic group of lava flows, tuffs, and interflow sediments that comprise part of the western limb of the Keweenawan syncline. The dip of the Duluth gabbro and the North Shore volcanic group is to the southeast under Lake Superior. Figure 1 shows the geographic location of the Duluth gabbro and the regional geology.

The geology and petrology of the rocks of the Duluth gabbro are well known from the studies of *Grout* [1918*a*, *b*, *c*] and *Taylor* [1964]. This report is concerned only with age determinations of the rocks of the Duluth gabbro complex. A review of the internal structure and petrogenesis of this complex will therefore be omitted.

Several sills of diabase and granophyre occur within the North Shore volcanic group stratigraphically above the Duluth gabbro complex. These are the Endion sill, the Northland sill, and the Lester River sill. The Endion sill is well exposed along the shore of Lake Superior in Duluth and has been studied by *Schwartz and Sandberg* [1940] as well as by *Ernst* [1960]. It consists of diabase that grades upward through a zone of intermediate rock into granophyre. The intermediate rock and granophyre together constitute about 40% of the sill in its exposure along the shore of Lake Superior. *Ernst* [1960] suggested that the salic fractions formed by differentiation of basaltic magma and then flowed up the dip and accumulated in their present position. The Endion sill is therefore also composite in nature.

Goldich et al. [1966] have recently reviewed the available age determinations of the Keweenawan rocks of the mid-continent region and concluded that the igneous activity occurred about 1100 m.y. ago within the period from 1000 to 1200 m.y. *Chaudhuri and Faure* [1967, 1968] reported age determinations of the igneous and sedimentary rocks of Keweenawan age on the Keweenaw peninsula of Michigan.

We report here Rb-Sr, whole-rock age determinations of the Duluth gabbro and of the related Endion sill of Duluth, Minnesota. The Duluth gabbro has already been dated by the K-Ar method by *Goldich et al.* [1961] and by the U-Pb method applied to zircons by *Silver and Green* [1963]. The Duluth gabbro complex is therefore among the most thoroughly dated igneous intrusives and provides an opportunity to re-evaluate the decay constant of Rb^{87} by comparison with the U-Pb decay schemes.

ANALYTICAL PROCEDURES AND RESULTS

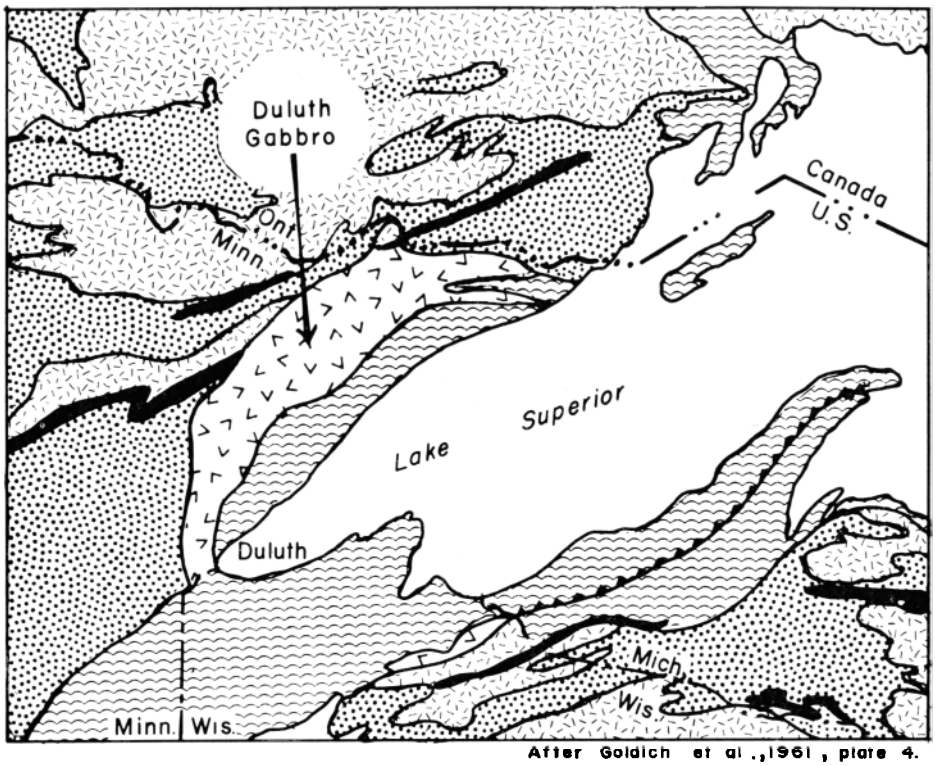
The specimens from the Duluth gabbro complex used in this study were collected along Skyline Drive in the Enger Tower area of Duluth. Petrographic descriptions of these samples are provided in Table 1. The Endion sill was sampled along the shore of Lake Superior in Duluth by using a geologic sketch map published by *Ernst* [1960]. Samples were collected at the same localities from which Ernst had collected earlier; these are also described in Table 1.

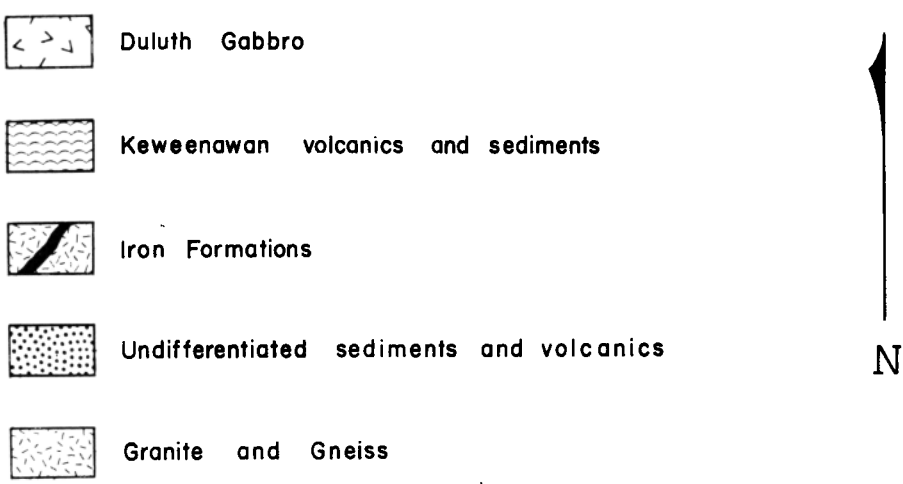
Concentrations of rubidium and strontium were measured by isotope dilution by using spikes enriched in Rb^{87} and Sr^{86}, respectively.

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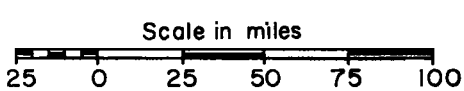


Fig. 1. Regional geology of the Duluth gabbro complex in northern Minnesota.

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TABLE 1. Modal Analyses of Rock Specimens from the Duluth Gabbro Complex and the Endion Sill*

Sample Number	Rock Type	Alk. Felds.	Plag.	Comp.	Qtz.	Hbld.	Pyr.	Ol.	Mag.	Chlorite	Others	Remarks
Duluth Gabbro Complex												
R4366	Olivine Gabbro	...	78	$\sim An_{58}$	Tr.	Tr.	5	13	3	...	biotite 1%, bowlingite 1%	Skyline Drive, Enger Tower, Duluth, Minn.
R1231	Gabbro				...	No handspecimen available				...		Duluth Gabbro complex.
R4362	Anorthositic gabbro	...	83	$\sim An_{45}$	Tr.	9	1	...	2	5	biotite, apatite	Skyline Drive, Enger Tower, Duluth, Minn.
R4365	Granophyre (granite)	60	Tr.	$\sim An_{23}$	9	...	...	...	...	3	qtz-Kspar micrograph. intergr. 29% apatite, rutile	Skyline Drive, Enger Tower, Duluth, Minn.
R44361	Granophyre (granite)	59	9	$\sim An_{40}$	12	12	Tr.	...	5	2	apatite,	Skyline Drive, Enger Tower, Duluth, Minn.
R4364	Granophyre (granite)	64	3	$\sim An_{34}$	7	10	Tr.	...	11	3	calcite 2%, biotite, apatite, zircon	Skyline Drive, Enger Tower, Duluth, Minn.
Endion Sill												
R4369	Diabase	...	44	$\sim An_{55}$	...	1	26	...	12	15	apatite 2%, biotite	Shore of Lake Superior, Duluth, Minn. Same location as 2, B of *Ernst* [1960, p. 288].
R4371	Interm. rock	12	39	$\sim An_{49}$	7	Tr.	12	...	9	19	apatite 3%, biotite	Same location as M3265 of *Ernst* [1960, p. 288].
R4372	Interm. rock	57	3	?	9	...	Tr.	...	6	25	rutile, biotite, apatite 1%	Same location as 8 of *Ernst* [1960, p. 288].
R4373	Granophyre	59	Tr.	...	16	...	...	...	5	20	calcite and apatite	Same location as 8 C of *Ernst* [1960, p. 288].

* In units of volume per cent.

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The isotope composition of strontium was determined on separate unspiked portions of each sample. Six of the 11 rock samples were analyzed in duplicate. All measurements were made on mass spectrometers having 6-inch radius, 60°-curvature analyzing tubes. Ion beams were generated by surface ionization from single tantalum filaments that were precleaned in the mass spectrometer before each measurement. The analyses were carried out in the geochronology laboratories of the Massachusetts Institute of Technology, Kansas State University, and Ohio State University. The Eimer and Amend $SrCO_3$ (lot 492327) was analyzed repeatedly in all three laboratories with excellent agreement. The data are compiled in the note to Table 2.

Age determinations for the Duluth gabbro complex and the Endion sill were made from the data in Table 2 by the isochron method using a value of 1.39×10^{-11} yr $^{-1}$ for the decay constant of Rb^{87}. The slopes and zero intercepts of the best straight lines were calculated by regression of x on y and y on x. The error estimates of the ages were calculated from one standard deviation of the slopes of the isochrons; these are presented in Figures 2 and 3.

Both rock suites form satisfactory isochrons from which the following dates can be derived: Duluth gabbro, 1115 ± 14 m.y.; Endion sill, 1092 ± 15 m.y. The two dates differ by only 23 m.y. Although the Endion sill appears to be the younger of the two intrusives, the difference in age is not statistically significant. The initial Sr^{87}/Sr^{86} ratio of the Duluth gabbro is 0.7055 ± 0.0003 whereas that of the Endion sill is 0.7046 ± 0.0006. Both values are somewhat higher than would be expected for magma derived exclusively from the upper mantle and suggest that some assimilation of crustal rocks has occurred. The difference in the initial Sr^{87}/Sr^{86} ratios is probably not statistically significant but suggests that small inhomogeneities in the isotope composition of strontium in the magma may have existed as a result of assimilation of crustal rocks by the magma.

TABLE 2. Analytical Data for Total-Rock Samples of the Duluth Gabbro and the Endion Sill

Sample Number	Description	Rb, ppm	Total Sr, ppm	$\left[\frac{Rb^{87}}{Sr^{86}}\right]_{atomic}$	$\left[\frac{Sr^{87}}{Sr^{86}}\right]^{*}_{corr.}$
		Duluth Gabbro Complex			
R4366	Olivine gabbro	7.96	343.9	0.067	0.7063
	Layered series	7.01	324.9	0.062	0.7054
R1231	Gabbro	6.80	345.7	0.057	0.7066
R4362	Anorthositic gabbro	31.5	402.4	0.227	0.7091
R4365	Granophyre	5.3	36.3	0.423	0.7121
R4361	Granophyre	113.5	305.8	1.076	0.7225
		112.2	312.1	1.042	0.7229
R4364 A†	Granophyre	97.6	120.2	2.359	0.7425
R4364 B	Granophyre	99.4	118.0	2.447	0.7432
		Endion Sill			
R4369	Diabase	40.9	215.7	0.548	0.7121
		40.9	215.4	0.550	0.7120
R4371	Intermediate rock	52.0	192.5	0.782	0.7175
		49.9	188.1	0.778	0.7170
R4372	Granophyre	125.1	...	...	0.7400
		125.1	158.1	2.29	0.7398
R4373	Granophyre	151.5	84.9	5.20	0.7826
		147.1	83.0	5.12	0.7840

The Eimer and Amend, $SrCO_3$ isotope standard, lot 492327, was analyzed repeatedly in the course of this study in all three laboratories with the following results: O.S.U. $(Sr^{87}/Sr^{86})_{corr} = 0.7083 \pm 0.0003$, M.I.T. $(Sr^{87}/Sr^{86})_{corr} = 0.7081 \pm 0.0003$, K.S.U. $(Sr^{87}/Sr^{86})_{corr} = 0.7082 \pm 0.0004$.

* Corrected for isotope fractionation to a standard value of the Sr^{86}/Sr^{88} ratio of 0.1194.

† R4364 A and B are two different powders prepared from the same hand specimen.

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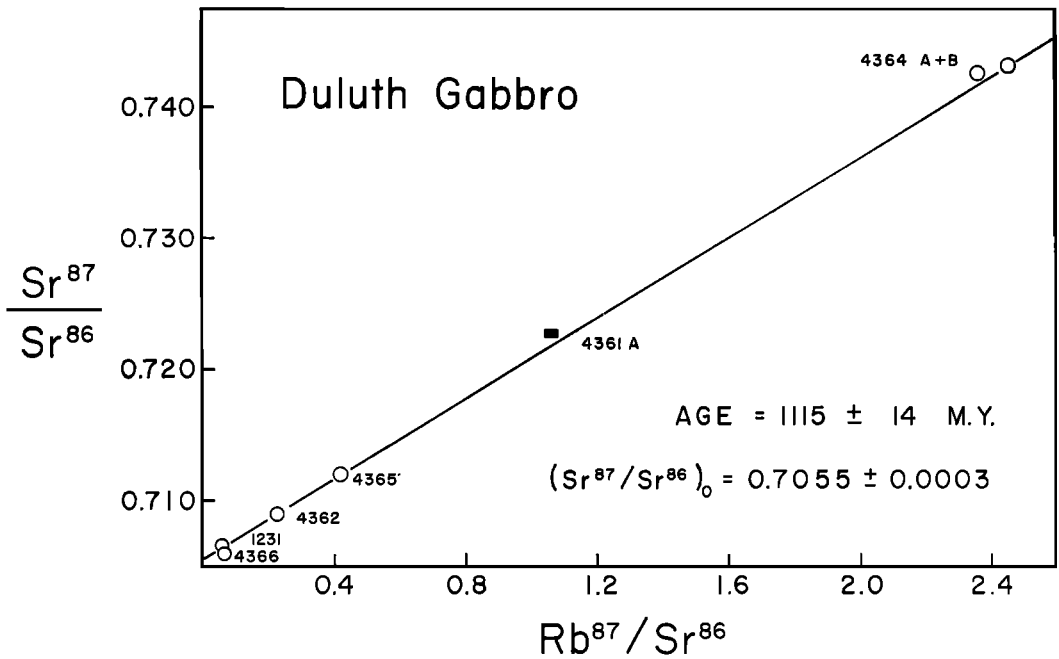


Fig. 2. Strontium evolution diagram of a suite of rocks from the Duluth gabbro complex at Duluth, Minnesota.

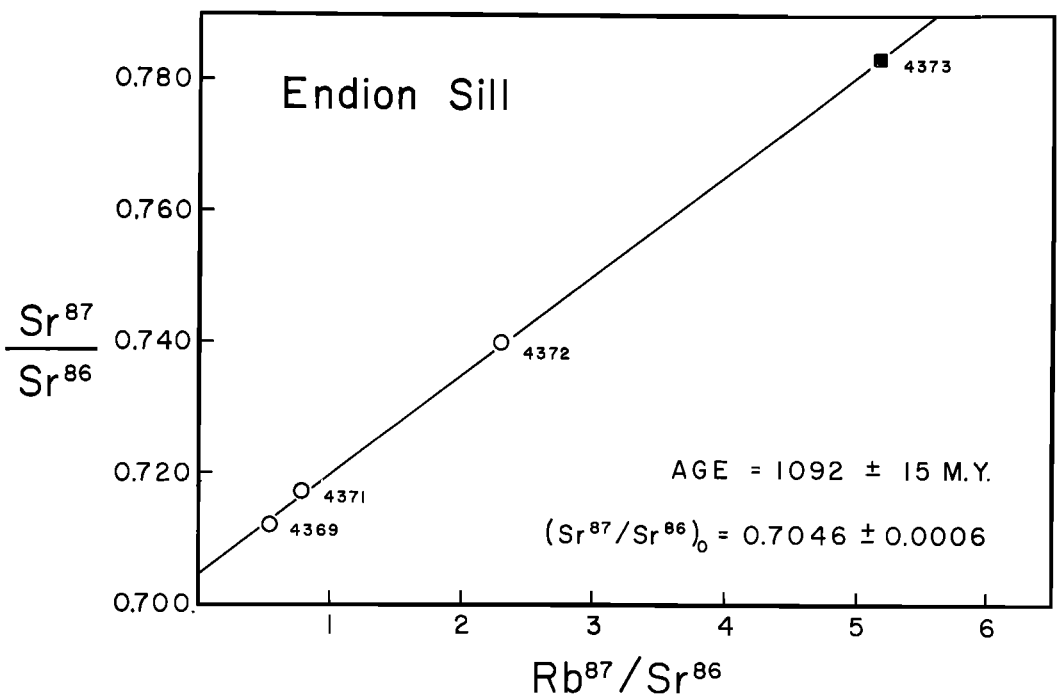


Fig. 3. Strontium evolution diagram of a suite of rocks from the Endion sill collected along the shore of Lake Superior at Duluth, Minnesota.

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Comparison of Age Determinations of Duluth Gabbro

Silver and Green [1963] dated zircons extracted from a rhyolite flow in the lower part of the North Shore volcanic group, from ferrogranodiorite and granophyre of the Duluth complex and the Endion sill, and from the granite which intrudes the Mellen gabbro at Mellen, Wisconsin. These zircons therefore span the period of igneous activity which occurred in this area. The U-Pb systems of these zircons form a linear array in a 'Concordia' diagram and indicate a collective age of 1115 ± 15 m.y. for these zircons. The agreement of the U-Pb zircon date and our Rb-Sr isochron date for the Duluth gabbro is excellent. This agreement suggests that all of the rocks sampled by *Silver and Green* [1963] crystallized in a relatively short period of time 1115 ± 15 m.y. ago. The concordance of the U-Pb zircon date and the Rb-Sr isochron date of the Duluth gabbro supports a value of 1.39×10^{-11} yr^{-1} for the decay constant of Rb^{87} which was derived originally by *Aldrich et al.* [1956] from a comparison of concordant U-Pb ages of uraninites and monazite with $*Sr^{87}/Rb^{87}$ ratios of coexisting micas in six pegmatites ranging in age from 2640 ± 100 to 375 ± 10 m.y. The change in value of the decay constant of Rb^{87} to 1.43×10^{-11} yr^{-1} suggested by *McDougall et al.* [1966] is not confirmed by this study.

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